

Spectral Compression as a Design Parameter for Multimodal Waste Segmentation

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Abstract—Hyperspectral imaging separates plastics that look identical in RGB, but a $256 \times 256 \times 224$ cube costs 29.4 MB per frame, which is too much to move across a sorting line in real time. Principal component analysis (PCA) reduces the cube to k channels before transmission, so k sets both the link budget and the achievable accuracy. Published work fixes $k = 3$ and justifies the choice by retained variance. We test that justification. We train a CMX-B0 hybrid-fusion network on the SpectralWaste dataset with RGB and PCA-reduced hyperspectral inputs and sweep k from 1 to 12. Accuracy does not saturate at $k = 3$: the first three components retain 99.595% of the training-set spectral variance, yet raising k from 3 to 12 adds 2.35 mIoU points, a 6.6% relative gain from components that together carry at most 0.405% of the variance. Retained variance is a poor predictor of segmentation accuracy. The curve flattens at $k = 6$, where the spectral payload is still $18.7\times$ smaller than the raw cube, and we would select that operating point over the published $k = 3$. Our reproduction reaches 35.50% mIoU against 56.6% published for the same architecture and modality; the gap persists at $k = 12$, which rules out compression depth and points instead to training from random initialisation. Per-class IoU tracks mean object size rather than class frequency: the four classes above 4000 pixels per instance reach 41–62% IoU, while *filament* at 606 pixels per instance collapses to 0.4%. We also find that the network marks material in frames whose ground truth is empty, at a confidence indistinguishable from its correct predictions.

Index Terms—hyperspectral imaging, semantic segmentation, multimodal fusion, dimensionality reduction, waste sorting

I. INTRODUCTION

Recycling facilities sort waste on moving conveyor belts under clutter, occlusion and deformation. Manual sorting exposes workers to sharp and contaminated material and produces inconsistent results. Automated sorters based on RGB cameras and convolutional networks handle shape and colour well, but they cannot separate polymers that share an appearance. A transparent polyethylene film and a transparent polypropylene film differ chemically and are visually identical. The limitation is visible in the taxonomies of RGB waste benchmarks: ZeroWaste [8], collected from a real sorting plant, separates soft from rigid plastic, a distinction defined by thickness and

rigidity rather than by appearance, and one that a top-view colour image cannot resolve directly.

Hyperspectral imaging (HSI) addresses this. A short-wave infrared sensor samples reflectance across hundreds of contiguous bands, and polymer families have distinct absorption features between 900 and 1700 nm [1]. The cost is data volume. The SpectralWaste sensor stores cubes of $600 \times 640 \times 224$ at 16-bit precision, 172 MB per frame. Even the downsampled 256×256 version used for training is 29.4 MB against 0.20 MB for the matching RGB frame, a factor of 149.

The standard answer is PCA. Reducing the spectral axis to k components before the network shrinks the payload and lets a hyperspectral stream reuse an RGB-shaped encoder. Casao *et al.* [1] adopt $k = 3$, report that the reduced input performs close to the full cube, and attribute that result to the high variance captured by three components. They do not report intermediate values of k , so the accuracy–bandwidth curve between $k = 3$ and $k = 224$ is unmeasured.

Research gap. Two questions remain open. First, k fixes the uplink budget of a distributed sorting station, so it is a system design parameter, yet no published measurement relates it to segmentation quality on this dataset. The justification offered for $k = 3$ is retained variance, and whether that quantity predicts accuracy has not been tested. Second, CMX-style fusion networks are normally initialised from ImageNet-pretrained transformer backbones. Whether the reported accuracy survives training from scratch matters to anyone deploying on a private industrial dataset with no comparable pretraining corpus.

Contributions.

- 1) We measure segmentation accuracy as a function of the number of retained principal components, $k \in \{1, 2, 3, 6, 12\}$, against transmitted payload per frame, and show that retained variance does not predict accuracy: components carrying under 0.405% of the variance add 2.35 mIoU points (Section V).
- 2) We reproduce CMX-B0 with RGB and PCA-3 from random initialisation, quantify the gap against the pub-

lished figure, and show that it persists at $k = 12$, which isolates initialisation rather than spectral compression as the cause.

- 3) We report a failure mode that aggregate mIoU hides: the network marks material in frames whose ground truth is empty, at confidence values inside the range it produces on correct predictions.

II. SYSTEM MODEL

A. Sensing Configuration and Assumptions

We assume the acquisition geometry of the SpectralWaste facility [1]: two synchronised line-scan cameras, a Teledyne DALSA Linea RGB sensor and a Specim FX17 hyperspectral sensor covering 224 contiguous bands from 900 to 1700 nm, mounted 1.7 m above a conveyor belt inside an industrial enclosure with LED and infrared halogen illumination.

Three assumptions follow, and each constrains what the system can claim.

A1: The two modalities are spatially registered. Both streams are cropped to the common field of view and re-sampled to 256×256 . Registration is not exact, because the cameras are line-scan devices at different positions and standard pinhole calibration does not apply [1]. Residual misalignment is one reason hybrid fusion outperforms early fusion, as Section III shows.

A2: Illumination is controlled. The enclosure fixes the light source, so raw digital numbers are treated as a stable material signature without radiometric calibration to reflectance. A model trained this way is tied to that installation.

A3: Batch inspection, not belt-rate control. We target an auditing station that reports material composition per batch. This admits a server-side model and a per-frame budget of a few hundred milliseconds, rather than the sub-30 ms budget of an actuator-in-the-loop ejector.

B. Problem Formulation

Let $\mathbf{X}_{\text{rgb}} \in \mathbb{R}^{H \times W \times 3}$ be the RGB image and $\mathbf{X}_{\text{hsi}} \in \mathbb{R}^{H \times W \times \Lambda}$ the co-registered cube, with $H = W = 256$ and $\Lambda = 224$. Let $\mathbf{x}(u, v) \in \mathbb{R}^{\Lambda}$ denote the spectrum at pixel (u, v) . We seek a map

$$f_{\theta} : \mathbb{R}^{H \times W \times 3} \times \mathbb{R}^{H \times W \times k} \rightarrow \{0, 1, \dots, 6\}^{H \times W}, \quad (1)$$

where class 0 is background and classes 1 to 6 are *film*, *basket*, *cardboard*, *video tape*, *filament* and *bag*. The second input is the compressed spectral tensor $\mathbf{Z} \in \mathbb{R}^{H \times W \times k}$ defined in Section IV-A.

C. Evaluation Metric

We accumulate a confusion matrix over all pixels of a split and compute

$$\text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{mIoU} = \frac{1}{6} \sum_{c=1}^6 \text{IoU}_c. \quad (2)$$

Background is excluded from the mean, following [1]. This matters: background occupies 81.20% of all pixels, and including it would raise our reported figure by 7.5 points

TABLE I
SPECTRALWASTE CLASS COMPOSITION. INSTANCE COUNTS FROM [1];
PIXEL STATISTICS MEASURED OVER ALL SPLITS, LOSS WEIGHTS OVER THE
TRAINING SPLIT. TEST IOU IS OUR CMX-B0 WITH RGB AND PCA-3.

Class	Inst.	Pixels (%)	Px/inst.	w_c	IoU (%)
background	—	81.20	—	0.6745	87.80
film	339	5.97	9 840	0.9393	56.49
basket	300	3.30	6 130	1.0000	41.95
cardboard	68	0.67	5 490	1.2128	61.54
video tape	287	1.12	2 180	1.1334	11.80
filament	111	0.12	606	1.5249	0.42
bag	954	7.63	4 460	0.9017	40.82

without saying anything about object detection. The statistic is accumulated over the dataset rather than averaged over images, so large objects dominate their class score.

D. Dataset

SpectralWaste [1] contains 852 annotated multimodal images with 2059 object instances, plus 6803 unlabelled pairs. Ground-truth masks were drawn on the RGB images with an interactive tool built on the Segment Anything Model [7]. We use the official split of 514 training, 167 validation and 171 test images, and supervise with the RGB-frame labels.

Table I shows the class composition, measured over all 852 masks (55.8 M pixels). The imbalance is severe, and instance count and pixel share disagree about which classes are rare. *Bag* contributes 954 of 2059 instances against 68 for *cardboard*, yet *cardboard* covers only 0.67% of pixels and *filament*, with 111 instances, covers 0.12%. The quantity that turns out to predict per-class accuracy is neither of these but the mean number of pixels per instance, which we report in the same table and return to in Section V.

III. BENCHMARK METHODS

We compare against the three architectures evaluated on this dataset in [1]. All operate at 256×256 and were selected there for low inference cost.

MiniNet-v2 [4] is a convolutional segmentation network using multi-dilation depthwise separable convolutions. It is the cheapest option at 0.52 M parameters and 1.34 GFLOPs, and reaches 44.5% mIoU on RGB alone.

SegFormer-B0 [3] is a hierarchical transformer with an all-MLP decoder. On RGB it reaches 48.4% mIoU at 3.72 M parameters.

Both are unimodal by design and are extended to two modalities by *early fusion*, concatenating the RGB and spectral channels before the first convolution. This is the natural baseline and it fails in an informative way. For MiniNet-v2, early fusion of RGB and PCA-3 gives 41.7% mIoU, below the 44.5% from RGB alone and the 49.0% from PCA-3 alone. SegFormer-B0 loses the same way against its spectral-only score: 48.5% fused against 53.5% from PCA-3 alone. Concatenating the two modalities at the input destroys information that either one carries separately. With imperfect registration a single early filter bank has to explain two spatially inconsistent signals at once.

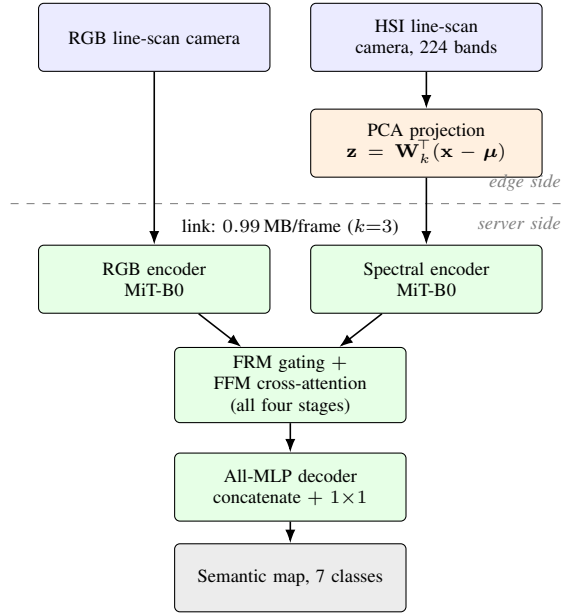


Fig. 1. Processing chain. The spectral cube is projected onto k principal components at the acquisition side, so the link carries k channels instead of 224. The server runs dual-stream encoding with feature rectification and cross-modal fusion at every stage. The quoted payload is the 8-bit RGB frame plus the $k = 3$ spectral tensor in FP32.

TABLE II

PUBLISHED RESULTS ON SPECTRALWASTE [1], MIOU OVER SIX OBJECT CLASSES. HYPER3 DENOTES PCA REDUCTION TO THREE COMPONENTS.

Backbone	Modality	mIoU (%)	Param. (M)	GFLOPs
MiniNet-v2	RGB	44.5	0.52	1.34
MiniNet-v2	HYPER3	49.0	0.52	1.43
MiniNet-v2	RGB+HYPER3	41.7	0.52	1.46
SegFormer-B0	RGB	48.4	3.72	3.51
SegFormer-B0	HYPER3	53.5	3.72	3.60
SegFormer-B0	RGB+HYPER3	48.5	3.72	3.63
CMX-B0	RGB+HYPER	58.2	11.54	8.37
CMX-B0	RGB+HYPER3	56.6	11.19	5.62

CMX-B0 [2] avoids this by keeping the streams separate and exchanging information between them at every stage. On this dataset it reaches 58.2% mIoU with the full cube and 56.6% with PCA-3, the two best results reported. Table II collects the published figures.

IV. PROPOSED SYSTEM

Fig. 1 shows the chain. The architecture follows [2] as adapted to this dataset in [1]; what we propose is the treatment of k as an explicit design variable of the acquisition-server split, and Section V measures the value it should take. We describe each stage as implemented, including the values that matter for reproduction.

A. Spectral Compression at the Acquisition Side

We treat every pixel spectrum as one sample. Let $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$ be all pixels of the 514 training cubes, so $N = 514 \times$

$256^2 \approx 3.37 \times 10^7$ samples in $\mathbb{R}^{224}$. With

$$\boldsymbol{\mu} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i, \quad \boldsymbol{\Sigma} = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^\top, \quad (3)$$

let $\mathbf{W}_k = [\mathbf{w}_1, \dots, \mathbf{w}_k] \in \mathbb{R}^{224 \times k}$ hold the eigenvectors of $\boldsymbol{\Sigma}$ for the k largest eigenvalues $\lambda_1 \geq \dots \geq \lambda_k$. Each pixel is projected as

$$\mathbf{z}(u, v) = \mathbf{W}_k^\top (\mathbf{x}(u, v) - \boldsymbol{\mu}) \in \mathbb{R}^k, \quad (4)$$

which stacks into $\mathbf{Z} \in \mathbb{R}^{H \times W \times k}$.

Three implementation choices are load-bearing.

Incremental estimation. The full training matrix is 30 GB in FP32 and does not fit in memory. We use incremental PCA [6], updating a rank- k SVD over mini-batches of ten cubes, so $\boldsymbol{\Sigma}$ is never formed explicitly.

Fit on the training split only. $\mathbf{W}_k$ and $\boldsymbol{\mu}$ are estimated on the 514 training cubes and applied unchanged to validation and test. Fitting on all splits would leak test statistics into the compression.

Input scale. We apply (4) to raw 16-bit digital numbers, whereas the reference pipeline rescales both modalities to $[0, 1]$ before reduction [1]. Scaling is linear, so the principal subspace and the retained-variance fractions are identical either way; only the magnitude of $\mathbf{Z}$ changes. Applying the projection to values 65 535 times larger makes our $\mathbf{Z}$ exactly 65 535 times the reference's, while both pipelines feed RGB in $[0, 1]$. The two streams have independent patch-embedding layers and each is followed by layer normalisation, so the scale is largely absorbed after the first block, but the gradient reaching that embedding is scaled with it. We return to this in Section V as a candidate contributor to the reproduction gap.

The first three components have measured variance ratios 0.9746, 0.0187 and 0.0026, retaining 99.595% in total, consistent with the 99.7% reported in [1]. On the usual reading this leaves almost nothing for further components to contribute. Section V shows that reading is wrong.

B. Dual-Stream Encoding with Feature Rectification

Both streams use a MiT-B0 hierarchical transformer [3] with embedding widths [32, 64, 160, 256], two blocks per stage, spatial-reduction ratios [8, 4, 2, 1] and stochastic depth 0.1. The streams share no weights.

At the output of stage s , let $\mathbf{F}_1, \mathbf{F}_2 \in \mathbb{R}^{C \times H_s \times W_s}$ be the RGB and spectral features. The Feature Rectify Module (FRM) [2] computes gates from the concatenation $\mathbf{U} = [\mathbf{F}_1; \mathbf{F}_2]$:

$$\mathbf{a} = \sigma(\text{MLP}[\text{GAP}(\mathbf{U}); \text{GMP}(\mathbf{U})]) \in \mathbb{R}^{2^C}, \quad (5)$$

$$\mathbf{b} = \sigma(\text{Conv}_{1 \times 1} \text{ReLU Conv}_{1 \times 1}(\mathbf{U})) \in \mathbb{R}^{2 \times H_s \times W_s}, \quad (6)$$

where σ is the logistic function and GAP, GMP are global average and max pooling. Splitting $\mathbf{a} = [\mathbf{a}^{(1)}; \mathbf{a}^{(2)}]$ and $\mathbf{b} = [\mathbf{b}^{(1)}; \mathbf{b}^{(2)}]$, rectification is a gated cross-injection:

$$\tilde{\mathbf{F}}_1 = \mathbf{F}_1 + \lambda_c \mathbf{a}^{(2)} \odot \mathbf{F}_2 + \lambda_s \mathbf{b}^{(2)} \odot \mathbf{F}_2, \quad (7)$$

$$\tilde{\mathbf{F}}_2 = \mathbf{F}_2 + \lambda_c \mathbf{a}^{(1)} \odot \mathbf{F}_1 + \lambda_s \mathbf{b}^{(1)} \odot \mathbf{F}_1, \quad (8)$$

with $\lambda_c = \lambda_s = 0.5$ and $\odot$ broadcasting elementwise. The gates are sigmoid-valued, not softmax-normalised: rectification is channel and spatial gating, not attention, and the two are not interchangeable. Each stream keeps its own residual path, so a noisy modality is attenuated rather than allowed to overwrite the other. Under assumption A1 that property is the reason to prefer this design over early fusion.

C. Cross-Modal Fusion

The Feature Fusion Module (FFM) merges the rectified features. Flatten each to tokens $\mathbf{T}_i \in \mathbb{R}^{N_s \times C}$ with $N_s = H_s W_s$, project and split,

$$[\mathbf{Y}_i, \mathbf{U}_i] = \text{split}(\text{ReLU}(\mathbf{T}_i \mathbf{P}_i)), \quad (9)$$

then apply cross-attention in its linear-complexity form. With $\mathbf{K}_i, \mathbf{V}_i$ linear projections of $\mathbf{U}_i$, per head of dimension d_h ,

$$\mathbf{C}_i = \text{softmax}\left(\frac{\mathbf{K}_i^\top \mathbf{V}_i}{\sqrt{d_h}}\right), \quad \tilde{\mathbf{U}}_1 = \mathbf{U}_1 \mathbf{C}_2, \quad \tilde{\mathbf{U}}_2 = \mathbf{U}_2 \mathbf{C}_1. \quad (10)$$

The softmax normalises over the rows of the $d_h \times d_h$ context matrix. This is not the standard $\text{softmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d})\mathbf{V}$ formulation, and the difference is what makes the module affordable: cost is $\mathcal{O}(N_s d_h^2)$ rather than $\mathcal{O}(N_s^2 d_h)$. At stage 1, $N_s = 64^2 = 4096$ and $d_h = 32$, so quadratic attention would cost 128 times more per head.

Each stream is updated residually, $\hat{\mathbf{T}}_i = \text{LN}(\mathbf{T}_i + [\mathbf{Y}_i, \tilde{\mathbf{U}}_i] \mathbf{Q}_i)$, and the two are concatenated and projected back to C channels by a $1 \times 1/\text{depthwise}/1 \times 1$ block with a parallel 1×1 residual. The key property of (10) is that each stream is queried against the *other* stream's context, so spectral evidence reaches the RGB stream and RGB boundaries reach the spectral stream at all four scales.

D. Decoder

The four fused maps are decoded with the all-MLP head of [3]. Each $\mathbf{F}_s$ is linearly projected to $d_e = 256$ channels, bilinearly resized to $H/4 \times W/4$, and the four are *concatenated*:

$$\mathbf{F}_{\text{agg}} = \text{ReLU BN Conv}_{1 \times 1}([\hat{\mathbf{F}}_1; \hat{\mathbf{F}}_2; \hat{\mathbf{F}}_3; \hat{\mathbf{F}}_4]). \quad (11)$$

Aggregation is concatenation followed by a learned 1×1 projection, not a weighted sum with scalar coefficients. The projection mixes all $4d_e$ channels, so the decoder combines scales per channel rather than per stage. Class logits come from a 1×1 convolution after dropout ($p = 0.1$) and are upsampled bilinearly to 256×256 .

E. Class-Imbalance Weighting

We minimise weighted cross-entropy over pixels Ω ,

$$\mathcal{L}(\theta) = -\frac{1}{|\Omega|} \sum_{p \in \Omega} \sum_{c=0}^6 w_c y_{p,c} \log \hat{y}_{p,c}, \quad w_c = \left(\frac{\text{med}_j f_j}{f_c}\right)^\gamma, \quad (12)$$

where f_c is the pixel frequency of class c in the training split and $\text{med}_j f_j$ its median over classes, here 3.30%, the

TABLE III
TRAINING CONFIGURATION. PER-CLASS LOSS WEIGHTS ARE GIVEN IN TABLE I.

Setting	Value
Optimiser	SGD, momentum 0.9, weight decay 5×10^{-4}
Learning rate	0.01, polynomial decay, power 0.9
Epochs / batch size	100 / 8
Initialisation	random (no ImageNet pretraining)
Loss	weighted cross-entropy (12), $\gamma = 0.12$
Input resolution	256×256
Encoder	MiT-B0 $\times 2$, widths [32, 64, 160, 256]
Decoder embedding	256
Parameters ($k=3$)	11.186 M
Training time	19 min per 100-epoch run

frequency of *basket*. Setting $\gamma = 1$ recovers median frequency balancing [5]; $\gamma \rightarrow 0$ recovers uniform weights.

We use $\gamma = 0.12$. The resulting weights span only 0.67 to 1.52 (Table I). For *filament* the frequency ratio $\text{med}_j f_j / f_c$ is 33.7, so $\gamma = 1$ would assign $w \approx 34$ while $\gamma = 0.12$ assigns 1.52. One exponent discards a factor of 22 in minority-class emphasis. Section V quantifies what that costs.

V. IMPLEMENTATION AND EVALUATION

A. Methodology

Environment. Kaggle notebooks, Ubuntu, Python 3.12, PyTorch 2.9.0 with CUDA 12.6 on a single NVIDIA Tesla T4 (15.6 GB). The runtime exposes two T4 cards; our pipeline is single-device and uses one. Spectral reduction uses scikit-learn `IncrementalPCA` and cube I/O uses `tiff`file. All timings are measured on that configuration.

Protocol. We use the official 514/167/171 split. PCA is fitted on the training split and applied to all three. Augmentation is random rotation of $\pm 30^\circ$ and horizontal and vertical flips, applied jointly to both modalities and the mask, matching [1]. No augmentation is used at validation or test. The sweep runs select the checkpoint by validation mIoU; the standalone PCA-3 run of Table V reports its final-epoch weights. Each result is computed once on the test split.

Sweep. For the k sweep we fit one PCA with 12 components and take the leading k channels for each run, so all runs share a single projection basis and differ only in how many components reach the network. Each configuration is trained from the same initialisation seed for 100 epochs. That basis is not the one used for the standalone PCA-3 model of Table V, which was fitted with $k = 3$, and it reports final-epoch rather than validation-selected weights. The two differences together move the result by 0.07 points, 35.43% against 35.50%, so we treat the runs as interchangeable: we use the sweep values when comparing across k and the standalone run when reporting per-class detail.

Table III lists the settings.

B. Results and Discussion

Retained variance does not predict accuracy. Fig. 2 and Table IV show mIoU rising from 31.05% at $k = 1$ to 37.78% at $k = 12$. The gain per added component falls monotonically,

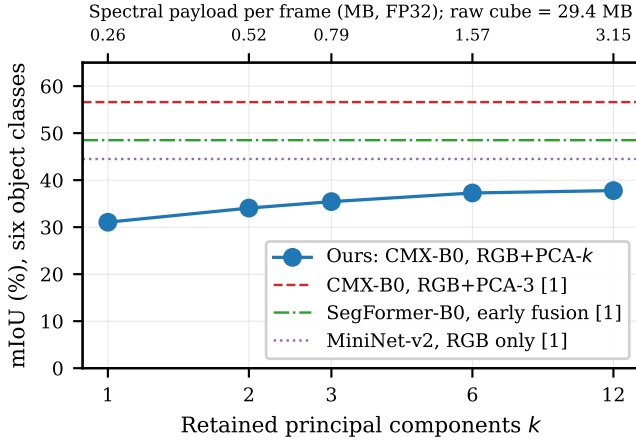


Fig. 2. Segmentation accuracy against the number of retained principal components. Solid: our CMX-B0 with RGB and PCA- k inputs, trained from random initialisation. Dashed horizontal lines: published references [1] for CMX-B0 with PCA-3, SegFormer-B0 with early fusion, and MiniNet-v2 on RGB. The upper axis gives the FP32 uplink payload per frame at each k .

TABLE IV

EFFECT OF THE NUMBER OF RETAINED PRINCIPAL COMPONENTS. PAYLOAD IS THE SPECTRAL STREAM ONLY, FP32, PER 256×256 FRAME; THE RAW CUBE IS 29.4 MB.

k	Payload (MB)	Compression	mIoU (%)	Δ mIoU
1	0.26	112.0 $\times$	31.05	–
2	0.52	56.0 $\times$	34.06	+3.01
3	0.79	37.3 $\times$	35.43	+1.37
6	1.57	18.7 $\times$	37.27	+1.84
12	3.15	9.3 $\times$	37.78	+0.51

and the curve flattens after $k = 6$: the step from $k = 6$ to $k = 12$ is +0.51 points, less than a third of the +1.84 gained from $k = 3$ to $k = 6$ and bought with twice as many extra channels.

The informative interval is the one the variance argument declares empty. The first three components carry 99.595% of the training-set spectral variance, so components 4 through 12 carry at most 0.405% between them. Those components add 2.35 mIoU points, a 6.6% relative improvement. A share of the variance below one part in 200 therefore produces a gain an order of magnitude larger in relative terms. Variance measures signal energy, not class separability, and a minority class with a distinctive but weak absorption feature contributes little energy by construction. Selecting k by a variance threshold optimises the wrong quantity.

The practical consequence is that $k = 3$ is not free. Moving to $k = 6$ doubles the payload, 1.57 MB per frame against 0.79 MB, and recovers 1.84 of the 2.35 available points. Under assumption A3 that trade is worth taking: at 1.57 MB the spectral stream is still 18.7 $\times$ smaller than the raw cube. We would select $k = 6$ for this installation rather than the published $k = 3$.

Reproduction gap. Our CMX-B0 with RGB and PCA-3 reaches 35.50% mIoU on test against 56.6% published for the

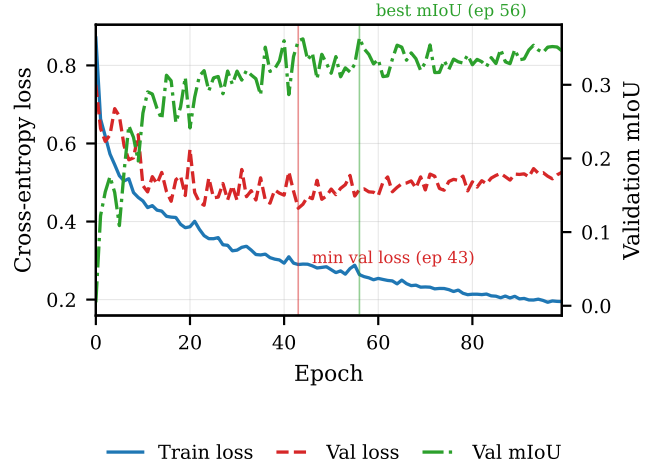


Fig. 3. Training dynamics over 100 epochs. Training loss decreases monotonically to 0.195 while validation loss reaches its minimum of 0.434 at epoch 43 and then rises. Validation mIoU peaks at 0.3636 at epoch 56.

TABLE V

PER-CLASS TEST IoU (%). OURS IS CMX-B0 WITH RGB AND PCA-3 TRAINED FROM RANDOM INITIALISATION; THE REFERENCE IS THE SAME CONFIGURATION IN [1].

Class	Ours	Reference	Δ
film	56.49	71.7	–15.2
basket	41.95	71.6	–29.6
cardboard	61.54	71.7	–10.2
video tape	11.80	27.8	–16.0
filament	0.42	37.7	–37.3
bag	40.82	59.4	–18.6
mIoU	35.50	56.6	–21.1

same architecture and modality configuration [1]. The sweep eliminates one candidate explanation directly: at $k = 12$ we still reach only 37.78%, 18.8 points below the published $k = 3$ result, so compression depth cannot account for the gap.

Four differences remain, in what we judge to be descending order of importance. First, initialisation. The reference CMX pipeline loads ImageNet-pretrained MiT weights into both encoders; we train from random initialisation. With 514 training images and 11.19 M parameters, a transformer encoder has no opportunity to learn generic low-level features from the task data alone. Second, schedule: 200 epochs with AdamW at 10^{-3} against our 100 epochs of SGD at 10^{-2} . Third, the input scaling of Section IV-A: our spectral branch enters the network at 65535 times the magnitude used in the reference pipeline. Fourth, the loss exponent $\gamma = 0.12$ discussed below.

The per-class pattern in Table V supports the first explanation. Our largest deficits are on *filament* (–37.3 points) and *basket* (–29.6), the classes that depend most on fine texture and thin structure, which is what pretrained early layers supply. *Cardboard* loses only 10.2 points, and it is the class most separable by coarse spectral signature.

Training dynamics. Fig. 3 shows the three curves. Training

loss falls from 0.870 to 0.195. Validation loss reaches 0.434 at epoch 43 and rises afterwards while training loss keeps falling, so the model overfits from roughly epoch 43. Validation mIoU is noisier and peaks later, at 0.3636 in epoch 56. The two selection criteria therefore disagree by 13 epochs, and we select on mIoU because it is the reported metric. The early onset of overfitting with 514 images is consistent with the initialisation argument above.

Per-class behaviour. Neither instance count nor total pixel share explains Table V. *Bag* has the most instances and the largest pixel share yet ranks fourth of six by IoU, while *cardboard* has the fewest instances and the second-smallest pixel share yet ranks first. The quantity that orders the classes is the mean number of pixels per instance (Table I): the four classes above 4000 pixels per instance score between 40.8% and 61.5% IoU, while *video tape* at 2180 falls to 11.8% and *filament* at 606 to 0.42%. *Cardboard* is the one class that outranks its size, which is consistent with cellulose having a spectral signature well separated from the polymers around it.

Filament is effectively undetected. Its objects are a few pixels wide, and the decoder aggregates at $H/4$ before upsampling, so structures thinner than four pixels are not represented at the resolution where classification happens. The loss weighting compounds this. A class contributes to the gradient in proportion to $w_c f_c$, so with $w = 1.52$ and $f = 0.12\%$, *filament* accounts for 0.25% of the weighted pixel mass against 75% for background. Predicting filament as background is close to free under (12), and the model does exactly that. The difficulty is not ours alone: the reference reports its two lowest scores on the same two thin classes, 27.8% for video tape and 37.7% for filament, against 71.6–71.7% on the three large ones.

Fig. 4(b) shows that the failure is not a simple miss. On a frame whose ground truth is 10.92% *filament*, the model predicts 2.66% of the frame as *video tape* and places it along the same coiled hose. Both classes are thin elongated objects. The network resolves the shape and assigns the wrong material, so what is lost is the spectral discrimination between the two, not the spatial evidence.

Predictions on unannotated frames. Fig. 4(c) shows a behaviour that mIoU does not expose. The ground truth for this test frame is empty, yet the model labels 14.80% of the area as *bag* at 0.944 mean confidence. The regions it selects contain physical objects, including a crumpled sheet at the upper left with the appearance of plastic film.

Two explanations fit and our data cannot separate them. The objects may fall outside the six-class taxonomy, in which case background is the correct label and the model is wrong. Or the annotation may be incomplete for this frame, in which case the model is right and the metric is wrong. The system-level consequence is the same either way: a station reporting material composition per batch would over-report recoverable material here.

What the case does establish is that confidence carries no information about correctness in this regime. The correctly segmented frames in our test scan return mean confidences between 0.92 and 0.98, so 0.944 sits inside that range and no

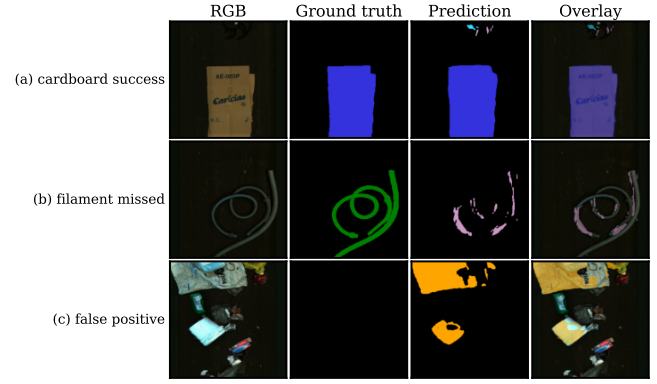


Fig. 4. Test-set predictions. (a) A large *cardboard* instance segmented cleanly, 0.967 class IoU. (b) A frame whose ground truth is 10.92% *filament*; the model resolves the shape but labels it *video tape*, giving 0 IoU. (c) A frame with empty ground truth in which the model marks 14.80% of the area as *bag* at 0.944 mean confidence. Columns: RGB input, ground truth, prediction, overlay.

threshold separates the two populations. A calibrated rejection mechanism, or an explicit *other waste* class, is needed instead. The 6803 unlabelled images in SpectralWaste are the obvious material for training one.

Deployment implications. At $k = 3$ the uplink carries 0.79 MB per frame for the spectral stream and 0.20 MB for RGB. On a 100 Mbit/s industrial link that is 79 ms of transmission per frame, so the five-second batch budget of assumption A3 is dominated by acquisition and inference rather than by the link. At $k = 12$ transmission rises to 267 ms, still a factor of 19 inside the budget. Bandwidth is not the binding constraint at this frame rate. Combined with Fig. 2, this argues for choosing k on accuracy grounds and for treating $k = 3$ as a lower bound rather than a default.

VI. CONCLUSION

We measured how spectral compression depth affects multi-modal waste segmentation on SpectralWaste. The main result is negative for the standard justification of $k = 3$: three components retain 99.595% of the training-set spectral variance, yet raising k from 3 to 12 adds 2.35 mIoU points, with the full sweep running from 31.05% to 37.78%. Components carrying under 0.405% of the variance therefore carry a measurable share of the discriminative signal, and retained variance is the wrong criterion for selecting k . The curve flattens at $k = 6$, which we would adopt in place of $k = 3$: it recovers 1.84 of the 2.35 available points at 1.57 MB per frame, still $18.7\times$ smaller than the raw cube, and transmission stays inside a five-second batch budget by a factor of 19 even at $k = 12$.

Our reproduction of CMX-B0 with RGB and PCA-3 reaches 35.50% mIoU against 56.6% published. The gap persists at $k = 12$, so compression is not its cause. We attribute it primarily to training from random initialisation rather than from ImageNet-pretrained encoders, supported by the per-class pattern: the largest deficits fall on the classes that depend on fine texture. This is a useful negative result for anyone

planning to deploy such a model on a proprietary industrial dataset without a pretraining corpus.

Limitations. We evaluate one architecture at one input resolution, on data from a single facility under fixed illumination, with a model trained on 514 images. Each configuration was trained once, so we cannot separate the 0.51-point step from $k = 6$ to $k = 12$ from run-to-run variance. We did not retrain the baselines ourselves, so Table II carries the training differences of the original work, including its pretrained initialisation and its 200-epoch schedule. The edge-server split of Fig. 1 is a design: we measured payload sizes, not end-to-end latency on hardware.

Future work. Loading pretrained MiT-B0 weights is the highest-value change and requires no new data. Selecting components by class separability rather than by variance follows directly from our main result: a supervised criterion such as linear discriminant analysis would rank spectral axes by what distinguishes materials rather than by what varies most, and should reach the accuracy of $k = 6$ at a smaller k . Raising γ in (12) towards 1, or replacing cross-entropy with a Dice or focal objective, targets the thin-object collapse. An *other waste* class trained on the unlabelled subset would address the confident predictions on unannotated frames, which matter more for a material-auditing report than a few points of mIoU.

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